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Validation of a Composite Outcome Score for Assessing Return to Sports After Achilles Tendon Repair

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Background: The Ankle-GO is a composite outcome score based on 4 functional tests and 2 patient-reported outcome measures. It was initially described and validated for its ability to predict return to sports (RTS) at the same level of play after lateral ankle sprains.

Purpose/Hypothesis: The main aim of this study was to assess the psychometric properties of the Ankle-GO in patients after Achilles tendon repair. The second objective was to evaluate its ability to predict RTS at the same level of play. It was hypothesized that this composite outcome score is a valid and reliable tool that is able to discriminate and predict the level of RTS at 9 months after Achilles tendon repair.

Study Design: Case-control study; Level of evidence, 3.

Methods: This was a prospective multicenter study. Patients who were active in sports and underwent primary acute Achilles tendon repair between April 2021 and December 2022 were included. A control group comprising 30 participants with no history of lower limb injuries was also evaluated. After standard rehabilitation for the first 2 months, the Ankle-GO was administered at 6 and 9 months after repair by the same experienced physical therapist. During testing, the patients were asked whether they were able to practice their main sport again and, if so, whether at the same or a lower level of play. A comparison between control participants and patients at 9 months after surgery was performed.

Results: A total of 50 patients (24 women and 26 men), with a mean age of 38.3 ± 10.1 years, were included. No patients were lost to follow-up at 9 months. The Ankle-GO score at 6 months was 10.7 ± 4.8. At 9 months, the score had risen by 4.8 points. All participants scored above the minimum threshold. The minimum score achieved was 3 points (n = 5 [10%]). Only 1 patient (2%) obtained the maximum score (25 points). There was a significant difference in the Ankle-GO score between patients at 6 and 9 months and between control participants and patients at 9 months after Achilles tendon repair. Of the 50 patients included in the study, 14 (28%) returned to their preinjury level of sport at 9 months. The ability of the Ankle-GO score at 6 months to predict RTS at the preinjury level at 9 months was good (area under the curve = 0.71 [95% CI, 0.56-0.85]; $P < .01$). A Youden index of 0.42 was observed for a cut-off score of 8 points, which corresponds to a sensitivity of 81.8% and a specificity of 41.2%.

Conclusion: The Ankle-GO is a valid tool to evaluate and discriminate patients during the RTS continuum after Achilles tendon repair and to predict RTS at the same level at 9 months after surgery. This composite outcome score assesses ankle function and the patient's psychological readiness to avoid premature RTS and worsening of the injury. Further studies are needed to assess its value in identifying and predicting patients at risk of a recurrence.

Keywords: Achilles tendon rupture; Achilles tendon repair; return to sports; Ankle-GO; psychological readiness

An Achilles tendon rupture (ATR) is a serious injury, with an annual estimated incidence of 20 to 30 cases per 100,000 people per year.[29,43] The prevalence of ATRs has been increasing by a mean of 2.4% per year since 1980, notably in older patients associated with increased sports participation. Almost 90% of these injuries occur during a jump or acceleration in men aged between 30 and 40 years.[21,30] There has been a rise in their incidence since the COVID-19 pandemic, potentially related to the resumption of sports activity since the lockdown,[3,7] and an increased prevalence in female patients.[21,29]

There is a lack of consensus regarding the superiority of outcomes after surgical versus nonsurgical treatment.[5,14] Surgical repair reduces the risk of reruptures compared with nonsurgical repair; however, it may expose the

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patient to a greater risk of complications.[53] Surgical repair also offers greater plantarflexion strength recovery according to many studies[19,27,63] and is considered to be more effective in the resumption of sports in competitive athletes.[8] Athletic patients notably have less comorbidities and consequently are less likely to suffer complications such as infections or wound dehiscence. Additionally, minimally invasive surgery reduces these risks.[18,36,60]

Early rehabilitation has been shown to improve outcomes and reduce the risk of reruptures after nonsurgical and surgical treatment.[9,15,23,31,35,49,62] Prolonged immobilization is now clearly identified as a risk factor for reinjuries.[28,32] Because surgical treatment is mainly recommended for young patients engaged in sports, return to sports (RTS) is therefore a major concern after an ATR.[38]

One of the aims after the treatment of an ATR is to return patients to their preinjury state, including sports activity and performance. Zellers et al[61] reported that a mean RTS rate of 80% within 6 months can be expected. However, at 1 year after ATR surgery when a defined method of determining the level of play was used, barely half of the patients had returned to a similar level of activity.[6] Persisting muscle deficits[20,22,26,42] and psychological reasons, such as the loss of motivation[12] or the fear of reruptures,[55] have been postulated as contributing factors.

To enable RTS, all these aspects must be taken into account and evaluated to assess the patient's ability to resume physical activity. The Achilles tendon Total Rupture Score[39] has demonstrated its ability to predict RTS at 1 year after an ATR. The Ankle Ligament Reconstruction–Return to Sport after Injury (ALR-RSI) scale[1] evaluates the patient's psychological readiness for RTS after ankle ligament surgery. This scale has been validated after both ankle ligament surgery and ATR surgery.[54]

The Ankle-GO is a composite outcome score based on 4 functional tests and 2 patient-reported outcome measures. It was initially described and validated for its ability to predict RTS at the same level of play after lateral ankle sprains.[45] The main aim of this study was to assess the psychometric properties of the Ankle-GO in patients after ATR surgery. The second objective was to evaluate its ability to predict RTS at the same level of play. The hypothesis was that the Ankle-GO is a valid and reliable tool that is able to discriminate and predict the level of RTS at 9 months after an ATR.

METHODS

This prospective study was conducted at 3 centers. This study was approved by the institutional review board and the ethics committee of our institution (No. 00010835). The study was conducted in accordance with the ethical standards set out in the 1964 Declaration of Helsinki. Informed consent was obtained from all the patients in this study. The inclusion criteria were as follows: (1) patients who underwent operative repair within 2 weeks of the primary acute ATR between April 2021 and December 2022 and (2) patients who practiced sports before an ATR and wished to return after rehabilitation. A control group comprising 30 participants with no history of lower limb injuries was also evaluated. The exclusion criteria were as follows: (1) patients who did not practice sports before the injury, (2) patients suffering from a rerupture or chronic ATR, and (3) patients aged <18 years.

Data Collection

The following demographic data were obtained from patient interviews and patient medical records: age at the time of surgery, sex, body mass index (BMI), type and level of preinjury sport, and time to surgery.

Surgical Technique

All patients underwent primary open repair of the ATR using the same technique by trained orthopaedic surgeons.[25] Surgery was performed under spinal anesthesia. A preoperative dose of 2 g of cefotiam was intravenously infused. Patients were positioned prone with a thigh tourniquet. A 6- to 8-cm longitudinal incision was made along the medial edge of the Achilles tendon. After incision of the paratenon, the hematoma was removed, and both the proximal and distal stumps were debrided. The ankle was plantarflexed to draw the tendon ends closer. The ruptured ends of the tendon were then brought together using a No. 2 nonabsorbable suture in a Krackow stitch configuration with giftbox tensioning. The final repair construct was reinforced with interrupted sutures. The paratenon and subcutaneous tissue were closed with 2-0 absorbable sutures. After skin closure, a sterile dressing was applied. The patient was placed into a walker boot with heel wedges postoperatively on the day of surgery.

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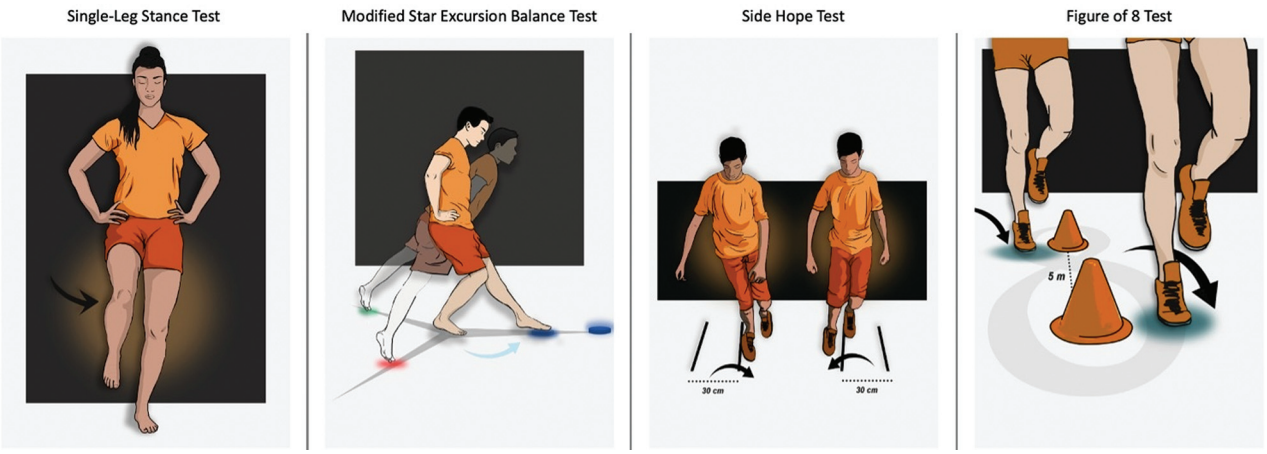

Figure 1. Functional tests of the Ankle-GO.

Rehabilitation

Patients underwent a standardized rehabilitation protocol for the first 2 months.[52] During the first 3 weeks, the operated ankle was immobilized in equinus in a walker boot with heel wedges. Nonweightbearing mobilization with crutches was permitted. Subsequently, partial progressive weightbearing was permitted between days 21 and 45, during which 1 heel wedge was removed per week, gradually reducing equinus. Patients underwent a standardized rehabilitation program that involved gradually increasing passive and active ankle range of motion; however, specific stretching and dorsiflexion were not allowed. The walker boot was discontinued after 45 days, and patients were then given an in-shoe heel wedge for a further 3 months and allowed to walk at full weightbearing. Subsequent calf strengthening exercises were performed.[51]

Ankle-GO

The Ankle-GO is a composite outcome score developed to assist in the decision-making process of RTS after a lateral ankle sprain. This tool combines 4 functional tests and 2 patient-reported outcome measures, of which 2 subscales of the Foot and Ankle Ability Measure (FAAM)[34] and the ALR-RSI scale have been validated for Achilles tendon abnormalities.[40,48,54] The 4 functional tests (Figure 1) used in the Ankle-GO are as follows:

- the single-leg stance test on a firm surface,[50]
- the modified version of the Star Excursion Balance Test,[41,46]
- the side hop test,[13] and
- the figure-of-8 hop test.[4]

The psychometric properties of these tests have been validated individually.[44] These 4 tests and the 2 questionnaires were selected and based on the recent consensus on RTS after lateral ankle sprains by the International Ankle Consortium: Pain severity, Ankle impairments, Athlete perception, Sensorimotor control, and Sport/functional performance.[56] The score ranges from 0 to 25, with the highest score indicating the best outcomes (Table 1).

The Ankle-GO was administered at 6 and 9 months after repair by the same experienced physical therapist. During testing, patients were specifically asked whether they had returned to their preinjury sport. The possible answers were "no," "yes, but not at the same level of play," and "yes, at the same or higher level."

The primary outcome of this study was the psychometric properties of the Ankle-GO in patients with ATRs. This included floor and ceiling effects, internal consistency, structural validity, and discriminant validity. The secondary outcome was the predictive ability of the Ankle-GO score at 6 months to determine RTS at 9 months.

Statistical Analysis

The Ankle-GO was conducted in adherence to the international COSMIN (COnsensus-based Standards for the selection of health Measurement INstruments) guidelines.[37] Means and standard deviations were used to describe quantitative variables. Numbers and percentages were used to describe dichotomous variables. The presence of floor or ceiling effects was noted if >15% of participants scored at the minimum (0 points) or maximum (25 points) threshold.

The internal consistency of the 7 components of the Ankle-GO was evaluated with the Cronbach alpha coefficient. This analysis was conducted on the total Ankle-GO score as well as each individual item. A Cronbach alpha between 0.7 and 0.9 suggests adequate internal consistency and minimal item redundancy.[37,57]

The Pearson correlation coefficient was calculated to assess the strength of the association of the individual components in relation to the others as well as of each component in relation to the total Ankle-GO score. The correlation was considered to be weak ($0.3 \geq r > 0.1$), moderate ($0.5 \geq r > 0.3$), or strong ($r > 0.5$).[47]

Discriminant validity was evaluated for all of the items as well as for the total Ankle-GO score using the Wilcoxon signed-rank test for paired samples between the results

TABLE 1
Ankle-GO Scoring[a]

	Result	Score	Maximum Score
Functional performance test			
SLS test	>3 errors	0	3
	1-3 errors	1	
	0 errors	2	
	No fear during test	+1	
mSEBT	<90%	0	7
	90%-95%	2	
	>95%	4	
	Anterior >60%	+1	
	Posteromedial >90%	+1	
	No fear during test	+1	
Side hop test	>13 s	0	5
	10-13 s	2	
	<10 s	4	
	No fear during test	+1	
Figure-of-8 hop test	>18 s	0	3
	13-18 s	1	
	<13 s	2	
	No fear during test	+1	
Patient-reported outcome measure			
FAAM			
Activities of Daily Living	<90%	0	4
	90%-95%	1	
	>95%	2	
Sports	<80%	0	
	80%-95%	1	
	>95%	2	
ALR-RSI scale	<55%	0	3
	55%-63%	1	
	64%-76%	2	
	>76%	3	

[a]ALR-RSI, Ankle Ligament Reconstruction–Return to Sport after Injury; FAAM, Foot and Ankle Ability Measure; mSEBT, modified Star Excursion Balance Test; SLS, single-leg stance.

obtained at 6 and 9 months. An independent *t* test was also used to compare patients at 9 months with the control group.

The predictive validity of RTS at the same or higher preinjury level was evaluated using the receiver operating characteristic curve based on the Ankle-GO score at 6 months and RTS status at 9 months. For this analysis, patients were dichotomized according to their RTS status between those who returned to their preinjury level or higher and those who did not (no RTS or lower level of sport). The area under the curve (AUC) was determined and considered to be null (AUC = 0.5), low ($0.5 < \text{AUC} < 0.7$), fair to good ($0.7 \leq \text{AUC} < 0.9$), high ($0.9 \leq \text{AUC} < 1$), or perfect (AUC = 1). The optimal cut-off score was determined with the Youden index (J = sensitivity + specificity – 1).

All statistical analyses were performed with SPSS software (Version 12.0; IBM). $P \leq .05$ was considered to be significant.

RESULTS

A total of 50 patients (24 female and 26 male), with a mean age of 38.3 ± 10.1 years, were included. No patients were lost to follow-up at 9 months. Overall, 4 patients were professional athletes (8%), while 24 practiced competitive sports (48%), 18 played recreational sports (36%), and 4 played occasional sports (8%). The demographic data of patients and control participants are shown in Table 2.

TABLE 2
Participant Characteristics[a]

	Patients (n = 50)	Controls (n = 30)	*P* Value
Sex, male/female, n	24/30	22/8	.249
Laterality, right/left, n	25/25	15/15	>.999
Age, y	38.3 ± 10.1	31.7 ± 13.5	<.001
BMI, kg/m^2	24.5 ± 2.6	22.4 ± 1.3	<.001
Type of sport			.414
Pivot and contact	26 (52.0)	9 (30.0)	
Pivot and no contact	13 (26.0)	14 (46.7)	
No pivot and no contact	11 (22.0)	7 (23.3)	
Level of sport			<.001
Professional	4 (8.0)	1 (3.3)	
Competitive	24 (48.0)	3 (10.0)	
Recreational	18 (36.0)	9 (30.0)	
Occasional	4 (8.0)	17 (56.7)	

[a]Data are shown as mean ± SD or n (%) unless otherwise indicated. BMI, body mass index.

TABLE 3
Functional Scores at 6 and 9 Months in the Patient Group[a]

	6 mo	9 mo	*P* Value
Ankle-GO	10.7 ± 4.8	15.5 ± 5.0	<.0001
FAAM, %			
Activities of Daily Living	88.5 ± 13.6	94.0 ± 12.0	<.0001
Sports	69.7 ± 19.6	84.3 ± 17.4	<.0001
ALR-RSI scale, %	60.1 ± 22.7	76.8 ± 21.1	<.0001
SLS test, errors	2.5 ± 2.2	1.7 ± 1.9	.0003
mSEBT, %			
Composite	81.1 ± 6.9	85.0 ± 7.7	<.0001
Anterior	59.3 ± 6.1	62.7 ± 6.6	<.0001
Posteromedial	93.5 ± 9.1	97.5 ± 9.9	.0005
Posterolateral	90.7 ± 9.9	94.9 ± 10.6	.0003
Side hop test, s	18.3 ± 10.6	14.1 ± 8.6	<.0001
Figure-of-8 hop test, s	17.6 ± 8.1	14.7 ± 6.6	<.0001

[a]Data are shown as mean ± SD. ALR-RSI, Ankle Ligament Reconstruction–Return to Sport after Injury; FAAM, Foot and Ankle Ability Measure; mSEBT, modified Star Excursion Balance Test; SLS, single-leg stance.

The Ankle-GO score at 6 months was 10.7 ± 4.8. At 9 months, the score had risen by 4.8 points (Table 3). Table 4 shows the evolution of the Ankle-GO score among professional/competitive athletes and recreational/occasional athletes.

No evidence of floor or ceiling effects was observed in the study. All participants scored above the minimum threshold. The minimum score achieved was 3 points (n = 5 [10%]). Only 1 patient (2%) obtained the maximum score (25 points).

The internal consistency of the Ankle-GO was good, with a Cronbach alpha coefficient of 0.78. Interitem

TABLE 4
Ankle-GO Score in Professional/Competitive Athletes and Recreational/Occasional Athletes[a]

	6 mo				9 mo			
	Same or Higher Level	Lower Level	No Sport	*P* Value	Same or Higher Level	Lower Level	No Sport	*P* Value
Professional and competitive	(n = 24) 11.9 ± 5.1	(n = 4) 12.5 ± 2.6	(n = 0) NA ± NA	.9474	(n = 6) 16.0 ± 5.9	(n = 14) 17.3 ± 4.5	(n = 8) 19.1 ± 2.5	.4736
Recreational and occasional	(n = 16) 7.9 ± 4.0	(n = 5) 13.0 ± 4.5	(n = 1) 9.0 ± NA	.0505	(n = 11) 11.4 ± 4.3	(n = 5) 12.6 ± 4.7	(n = 6) 15.7 ± 3.6	.1665

[a]Data are shown as mean ± SD. NA, not applicable because no patient returned to sports at the same level at 6 months.

TABLE 5
Results in Patients at 9 Months and in Control Group[a]

	Patients at 9 mo	Controls	*P* Value	Effect Size	95% CI
Ankle-GO	15.5 ± 5.0	19.6 ± 3.4	<.001	–0.926	–1.401 to −0.446
FAAM, %					
Activities of Daily Living	94.0 ± 12.0	99.8 ± 0.5	<.001	–0.603	–1.065 to −0.136
Sports	84.3 ± 17.4	98.9 ± 2.2	<.001	–1.061	–1.542 to −0.574
ALR-RSI scale, %	76.8 ± 21.1	96.1 ± 5.2	<.001	–1.136	–1.621 to −0.644
SLS test, errors	1.7 ± 1.9	1.2 ± 1.4	.174	0.318	–0.140 to 0.774
mSEBT, %					
Composite	85.0 ± 7.7	91.9 ± 6.7	.004	–0.695	–1.165 to −0.220
Anterior	62.7 ± 6.6	65.6 ± 5.0	.041	–0.487	–0.951 to −0.020
Posteromedial	97.5 ± 9.9	106.8 ± 11.2	<.001	–0.914	–1.392 to −0.429
Posterolateral	94.9 ± 10.6	91.9 ± 6.7	.162	0.331	–0.132 to 0.792
Side hop test, s	14.1 ± 8.6	11.6 ± 2.7	.125	0.359	–0.100 to 0.816
Figure-of-8 hop test, s	14.7 ± 6.6	11.7 ± 2.2	.022	0.543	0.079 to 1.004

[a]Data are shown as mean ± SD unless otherwise indicated. ALR-RSI, Ankle Ligament Reconstruction–Return to Sport after Injury; FAAM, Foot and Ankle Ability Measure; mSEBT, modified Star Excursion Balance Test; SLS, single-leg stance.

correlations were also high, with a mean of 0.85 (0.67-0.92, 95% CI). The deletion of items did not significantly decrease or increase the Cronbach alpha coefficient.

There was a significant difference in the Ankle-GO score between patients at 6 and 9 months and between control participants and patients at 9 months after the ATR (Tables 3 and 5).

Return to Preinjury Level of Sport or Higher

Of the 50 patients included in the study, 14 (28%) returned to their preinjury level of sport at 9 months (Table 6). The ability of the Ankle-GO score at 6 months to predict RTS at the preinjury level at 9 months was good (AUC = 0.71 [95% CI, 0.56-0.85]; $P < .01$). A Youden index of 0.42 was observed for a cut-off score of 8 points, which corresponds to a sensitivity of 81.8% and a specificity of 41.2% (Figures 2-4).

DISCUSSION

The most important finding of this study was the ability of the Ankle-GO score at 6 months to predict RTS at 9 months at the same preinjury level or higher. The Ankle-GO demonstrated excellent discriminant validity after ATRs, similar to that reported after lateral ankle sprains.[45] The tool was also responsive, and the score improved significantly between 6 and 9 months from 10.7 ± 4.8 to 15.5 ± 5.0. The same was true for all the components of the Ankle-GO. The internal consistency of the Ankle-GO was also optimal (0.7 < Cronbach alpha < 0.9), providing high interrelatedness without redundancy among items. No floor or ceiling effects were observed.

The ability of the 6-month Ankle-GO score to predict 9-month outcomes appears to be of significant interest to clinicians. Indeed, this late phase of the first postoperative year is regarded as highly critical in the literature.[33,58] For example, Trofa et al[58] reported that in professional players (National Basketball Association, National Football League, Major League Baseball, and National Hockey League) who sustained a primary complete ATR treated surgically, this injury prevented 30.6% of athletes from returning to their professional sports career. Moreover, those who did return played for shorter durations, participated in fewer games, and performed at a lower level than before their injury. This suggests that, according to the current definition of return to performance, some of these

TABLE 6
Results in Relation to RTS at 6 and 9 Months[a]

	6 mo				9 mo			
	Same or Higher Level (n = 1)	Lower Level (n = 9)	No Sport (n = 40)	*P* Value	Same or Higher Level (n = 14)	Lower Level (n = 19)	No Sport (n = 17)	*P* Value
Ankle-GO	9.0 ± NA	12.8 ± 3.6	10.3 ± 5.0	.3385	16.1 ± 4.9	16.1 ± 4.9	13.0 ± 5.3	.0440
FAAM, %								
Activities of Daily Living	100.0 ± NA	94.0 ± 5.7	87.0 ± 14.6	.0412	97.7 ± 3.7	95.9 ± 5.6	88.9 ± 18.8	.0842
Sports	93.8 ± NA	82.3 ± 11.2	66.3 ± 19.8	.0170	94.2 ± 5.9	84.9 ± 13.6	75.6 ± 22.9	.0098
ALR-RSI scale, %	65.0 ± NA	73.7 ± 19.8	56.8 ± 22.6	.1004	87.9 ± 14.6	76.3 ± 17.9	68.1 ± 25.3	.0246
SLS test, errors	7.0 ± NA	2.8 ± 3.0	2.4 ± 2.0	.3002	2.1 ± 2.1	1.6 ± 1.6	1.5 ± 1.9	.7643
mSEBT, %								
Composite	68.8 ± NA	84.3 ± 4.0	80.7 ± 7.1	.0820	86.9 ± 6.7	86.8 ± 6.9	81.6 ± 8.5	.0686
Anterior	58.9 ± NA	60.4 ± 6.0	59.1 ± 6.3	.9975	62.6 ± 5.3	64.6 ± 6.0	60.7 ± 7.9	.4235
Posteromedial	78.9 ± NA	95.8 ± 6.0	93.3 ± 9.5	.1963	100.5 ± 8.7	99.3 ± 8.8	93.0 ± 10.8	.0387
Posterolateral	68.4 ± NA	96.6 ± 6.2	89.9 ± 9.8	.0281	97.4 ± 11.4	96.6 ± 10.1	91.0 ± 9.9	.0588
Side hop test, s	35.0 ± NA	15.2 ± 8.0	18.6 ± 10.8	.2266	13.6 ± 6.5	12.4 ± 4.2	16.4 ± 12.8	.8994
Figure-of-8 hop test, s	23.0 ± NA	14.1 ± 1.9	18.3 ± 8.8	.3642	13.3 ± 4.6	13.1 ± 3.5	17.6 ± 9.5	.0576

[a]Data are shown as mean ± SD. ALR-RSI, Ankle Ligament Reconstruction–Return to Sport after Injury; FAAM, Foot and Ankle Ability Measure; mSEBT, modified Star Excursion Balance Test; NA, not applicable; RTS, return to sports; SLS, single-leg stance.

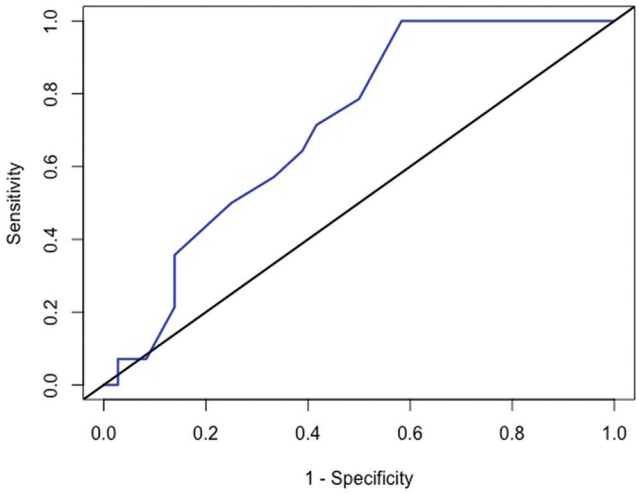

Figure 2. Receiver operating characteristic curve for the Ankle-GO score at 6 months for predicting return to preinjury level of sport or higher at 9 months.

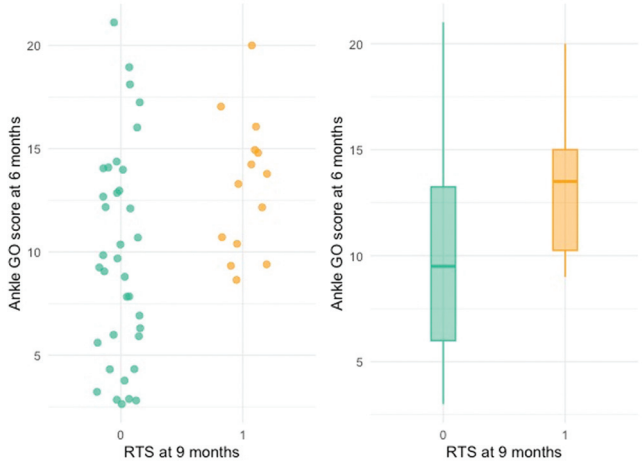

Figure 3. Scatter and box plots illustrating the relationship between the Ankle-GO score at 6 months and return to sports at 9 months (0: returned to the same level; 1: did not return to the same level).

athletes did not actually achieve proper return to performance.[2] However, these functional deficits were only observed at 1 year postoperatively compared with matched controls, indicating that players who resume play can expect to reach a performance level comparable with noninjured controls at 2 years after surgery. Therefore, a significant portion of recovery appears to occur during the period between 6 months and 1 year postoperatively. Utilizing a valid and quantifiable tool such as the Ankle-GO, particularly its relevance between 6 and 9 months, should greatly assist clinicians in refining rehabilitation during this critical phase to facilitate successful RTS and ultimately return to performance.

Given the prolonged recovery time after an ATR, it is not possible to compare RTS rates and Ankle-GO scores at 6 and 9 months with those after lateral ankle sprains.[45] The rates of RTS at the same level at 6 and 9 months in our study were 2% and 28%, respectively. Because of the differences in follow-up times between studies, it is difficult to compare with other series, with RTS rates of around 50% at 1 year.[6,38] The results, however, are different from the rate of 80% reported by Zellers et al.[61] In the subgroup of the meta-analysis by Zellers et al[61] in which RTS was clearly described, an RTS rate <77% was observed.

There are many reasons for this variability in results, including the intrinsic quality of the study, the type of

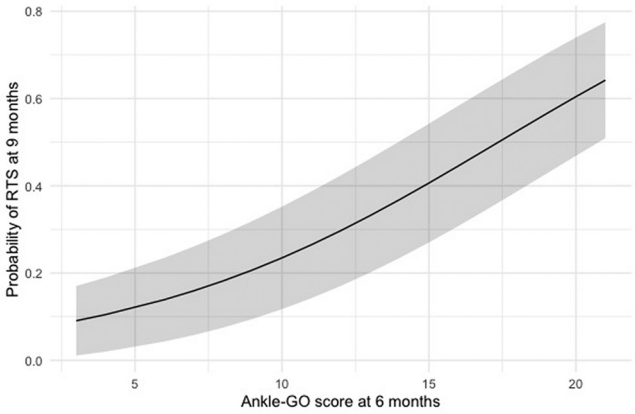

Figure 4. Probability of return to sports at 9 months based on the Ankle-GO score at 6 months.

treatment, the type of sport, and the level of sport, among others. The significant variability in RTS between series can be explained by several factors, including residual symptoms (such as pain, loss of muscle strength, and loss of endurance) and psychological factors (such as loss of motivation and fear of reruptures), and by the lack of a uniform definition of RTS.

The time to RTS after an ATR also varies widely between sports, and this is likely to be caused by the specific requirement for ankle plantarflexion strength within each sport.[59] Johns et al[24] reported a mean time to RTS of 11 months in 333 professional athletes. For those aiming to return to football, the elapsed time was 12.3 months, whereas for basketball, it was 10.5 months, and for soccer, it was 9.5 months.

The comparison between control participants and patients at 9 months after surgery showed significant differences, with poorer performance in the operated group in the total Ankle-GO score as well as all Ankle-GO items, except for the single-leg stance test, the posterolateral direction on the modified Star Excursion Balance Test, and the side hop test. This difference may also explain the low rate of RTS at the same level in our series.

In determining the criteria for RTS, several potential factors have been identified. This includes muscle strength, range of motion, endurance, functional recovery levels, pain levels, medical recommendations, psychosocial aspects, and anatomic and physiological properties of the Achilles tendon.[30] Many of these criteria, however, lack specific operational definitions, which complicates their validity and practical application.[16]

Hansen et al[17] showed that the Achilles tendon Total Rupture Score could be a predictive factor for RTS at 1 year but did not give a cut-off value. In our series, a cut-off score at 6 months for the ability to predict RTS at the same level at 9 months of 8 points was identified. Sensitivity and specificity were 81.8% and 41.2%, respectively. In comparison, in their study after ankle sprains, Picot et al[46] found an identical cut-off value but lower sensitivity (72%) and better specificity (66%).

Even if the scientific quality of the studies on ATRs is taken into perspective,[14] it has been shown that around 20% of patients who undergo ATR surgery will not RTS. In 75% of cases, the principal reasons for this is a fear of reinjury.[55] Psychological factors are important and should form a component of an RTS tool.[55]

This study has certain limitations. The test-retest reliability of the Ankle-GO has not been evaluated among patients after an ATR. Statistical components of the Ankle-GO were not measured, and test-retest reliability analysis was not performed. It was thought that it would be inappropriate to carry out these measurements, as retesting is generally conducted at 3-week intervals, but during rehabilitation, this period is likely to represent a significant change in performance. Fatigue will play a role if retesting is performed after a short period of time. The components of the Ankle-GO have been validated for reproducibility, including some that are specific to Achilles tendon injuries (eg, ALR-RSI scale), and the tool has already demonstrated excellent test-retest reliability in patients with lateral ankle sprains.[45] We do not believe that these results would differ significantly in patients who underwent ATR surgery. However, studies specifically evaluating test-retest reliability in this population are warranted, ideally with a 1- to 2-week interval between testing sessions.

Additionally, standard methods for monitoring ATRs[39] and the Achilles tendon resting angle[6] were not included in this study, as they are not components of the Ankle-GO, and their predictive value for RTS remains debated.[10,11,17] It could be interesting in the future to develop an even more specific tool for RTS after ATRs by integrating these 2 parameters.

It should be noted that the proportion of competitive athletes was significantly lower in the control group compared with the patient group. Additionally, patients were older and had a higher BMI compared with control participants. However, these differences do not fully explain the lower scores observed among patients, as all participants in both groups were aged between 30 and 40 years and fell within the same BMI category (18.5-24.9 kg/m^2 [normal weight]) according to the World Health Organization.

In this study, only RTS was evaluated; however, for many patients, time to return to work is also an important outcome measure to assist with treatment choice. This is a challenging outcome to determine, as the physical activity and Achilles tendon requirement for many occupations are very different.

CONCLUSION

The Ankle-GO is a valid tool to evaluate and discriminate patients during the RTS continuum after an ATR and to predict RTS at the same level of play at 9 months after surgery. This composite outcome score assesses ankle function and the patient's psychological readiness to avoid premature RTS and worsening of the injury. Further studies are needed to assess its value in identifying and predicting patients at risk of a recurrence.

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